PROBLEM

ARTIFICIAL INTELLIGENCE – CSA1717

ASSESSMENT TOOL 1 – CASE STUDY

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Course Code: CSA1717

Course Name: Artificial Intelligence

Assessment Title: Case Study

QUESTION 1: SMART MEDICAL DIAGNOSIS SYSTEM

Problem Statement

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system uses logical rules to identify possible diseases from patient symptoms.

Knowledge Base

Rule 1: If a patient has Fever AND Cough → Possible Flu.

Rule 2: If a patient has Fever AND Rash → Possible Measles.

Rule 3: If a patient has Cough AND Breathlessness → Possible Pneumonia.

Patient A Facts

- Fever = True
- Cough = True
- Breathlessness = True

Tasks

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm "Patient A has Pneumonia." Apply Backward Chaining and show the goal reduction steps to verify this goal.

QUESTION 2: AUTONOMOUS TRAFFIC MANAGEMENT AGENT

Problem Statement

A smart city traffic controller uses a **First-Order Logic (FOL)** based knowledge base to manage emergency vehicle routing.

Knowledge Base

Rule 1:

$\forall x$: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

Rule 2:

$\forall x$: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)

Rule 3:

$\forall x\ \forall y$: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)
- Vehicle(CarA)
- Behind(CarA, Ambulance)

Tasks

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that Proceed(CarA) is true. Convert the relevant rules into CNF and show the resolution steps.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are required and justify your answer logically.

QUESTION 3: AGRICULTURAL EXPERT REASONING SYSTEM

Problem Statement

An agricultural advisory logical agent operates with the following **propositional knowledge base**.

Knowledge Base

P1: SoilDry → IrrigationNeeded

P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod

P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Facts

- SoilDry = True
- CropWheat = True

Tasks

(a) Convert all sentences P1, P2, P3 and the given facts into Conjunctive Normal Form (CNF). Show the conversion steps.

(b) Using Resolution Refutation, prove the theorem ApplyDripMethod. Negate the goal, combine it with the knowledge base in CNF, and derive the empty clause □.

(c) If the fact SoilDry is removed from the knowledge base, analyze step-by-step how the logical conclusions change. Determine whether CropAtRisk still follows and justify your answer using appropriate reasoning.

QUESTION 4: WUMPUS WORLD KNOWLEDGE AGENT

Problem Statement

An AI agent navigating the **Wumpus World** perceives the following:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

The agent uses a propositional knowledge base with the following rules.

Knowledge Base

Rule 1:

Stench[1,2] → Wumpus is adjacent to cell [1,2]

Rule 2:

Breeze[1,1] → Pit is adjacent to cell [1,1]

Rule 3:

Glitter[x,y] → Gold[x,y]

Rule 4:

¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

Tasks

(a) Construct the agent's **belief state** from its percepts. Apply **Modus Ponens** systematically to derive new knowledge facts. List all derived conclusions.

(b) Using **Forward Chaining** from the agent's percepts, identify the possible locations of the **Wumpus and Pit**. Show each reasoning step precisely.

(c) Evaluate whether the path

[1,1] → [1,2] → [2,2]

is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion using logical inference.